

Green innovation and the CO₂ emissions gap: New evidence from OECD economies

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ABSTRACT

Carbon emissions are highly concentrated within societies; however, little is known about how technological change relates to this distribution. This study examines the relationship between directed technological innovation and intranational carbon inequality in OECD economies. Building on a framework that incorporates heterogeneous emitters, consumption-based footprints, and rebound effects, we distinguish between the scale and direction of innovation and employ the pooled Bewley estimator to identify long-run relationships in an unbalanced panel of OECD economies over 1990–2021, while accounting for slope heterogeneity and cross-sectional dependence. The results show that environment-related technological innovation is significantly associated with a narrower carbon emissions gap; a 1-percentage-point increase in the share of green technologies in domestic patenting corresponds to a 0.5%–1.4% lower level of carbon inequality. In contrast, aggregate patenting activity shows no systematic relationship with the emissions gap. Per-capita income follows an inverted-U pattern with the emissions gap, consistent with an Environmental Kuznets Curve interpretation, whereas urbanization is negatively linked to the gap, possibly reflecting diffusion and agglomeration channels. Trade openness shows a weak and inconclusive relationship. Overall, the findings suggest that the composition of technological progress, rather than its aggregate scale, is more closely linked to the distributional dimension of decarbonization. Policies that steer innovation toward low-carbon technologies may therefore contribute not only to emissions reduction but also to a more even distribution of emissions responsibilities in advanced economies.

1. Introduction

Climate change is increasingly recognized not only as an environmental challenge but also as a problem of inequality. A growing body of evidence indicates that carbon emissions are highly concentrated among a relatively small share of the population. Recent estimates based on consumption-based accounting indicate that in 2019, the richest 10% of the global population were responsible for nearly 48% of total emissions, while the poorest half contributed only approximately 12% (Chancel, 2022). Within this group, the wealthiest 1% alone accounted for approximately 17% of global emissions. Such disparities reflect structural differences in housing size, mobility patterns, ownership of energy-intensive durable goods, and access to high-carbon consumption, such as air travel (Ivanova et al., 2016; Oswald et al., 2020). These patterns imply that decarbonization is not only an environmental challenge but also a distributional one. Consequently, an important policy

question arises: Beyond reducing total emissions, how is technological change related to the distribution of carbon responsibility within countries?

Technological innovation is widely regarded as a central driver of long-term decarbonization. The literature on directed technical change emphasizes that policy incentives can steer research and development toward cleaner technologies, thereby altering the technological composition of production and energy use (Acemoglu et al., 2012). Empirical studies have shown that environmental regulation, carbon pricing, and renewable energy policies stimulate green patenting and accelerate technological development in low-carbon sectors (Johnstone et al., 2010; Aghion et al., 2016; Popp, 2019). A large body of empirical literature further demonstrates that environmentally oriented innovation is associated with lower carbon intensity and improvements in environmental performance across advanced economies (e.g., Fernández et al., 2018; Puertas and Marti, 2021; Kafeel et al., 2024).

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Despite this progress, the existing literature largely evaluates technological change through the lens of aggregate emissions or intensity measures. However, emission outcomes are inherently heterogeneous across households and income groups. Consumption-based accounting reveals that intra-national disparities often exceed cross-country differences, reflecting the concentrated carbon responsibility of high-income groups. While recent studies have examined the structural determinants of carbon inequality, including affluence, energy intensity, trade exposure, globalization, geopolitical risk, and macroeconomic conditions (Wiedmann et al., 2020; Oswald et al., 2020; Chen et al., 2024; Nguyen et al., 2025), the role of technological innovation remains insufficiently specified. In particular, existing studies do not distinguish between the scale of innovation and its environmental direction, although directed technical change theory suggests that these two dimensions may have different implications for emissions outcomes.

A key conceptual distinction, therefore, concerns the scale versus the direction of innovation. Aggregate inventive activity may raise productivity and improve efficiency, but it can also expand energy-service demand through rebound effects (Sorrell, 2009; Gillingham et al., 2016; Borenstein, 2015). By contrast, innovation directed toward low-carbon technologies may reduce the carbon intensity of consumption categories such as transport, housing, electricity use, and durable goods, which are central to unequal carbon footprints (Ivanova et al., 2016; Oswald et al., 2020). This distinction motivates the paper's main hypothesis: the environmental orientation of innovation may matter more than aggregate patenting activity for the long-run emissions gap between high- and low-income groups (hereafter, emitters).

Insights from macro-level emissions decomposition provide further motivation. The Kaya identity decomposes emissions into population, income, energy intensity, and carbon intensity components (York et al., 2003; González-Torres et al., 2021). Technological innovation primarily operates through the latter two channels. Recent studies incorporate patent-based indicators into such frameworks and show that environmental technologies contribute to decarbonization through structural shifts in the energy system (Lanzi et al., 2011; Costantini and Mazzanti, 2012). However, these analyses remain national in scope and do not assess whether reductions in intensity occur disproportionately among emitters.

A small but growing body of literature has begun to examine the distributional dimension of carbon emissions using panel econometric approaches. For example, Chen et al. (2024) show, for a panel of 38 developed and developing economies, that geopolitical risk, the capital-labor ratio, and per-capita income increase CO₂ emission inequality, whereas globalization mitigates it. Likewise, Nguyen et al. (2025) analyze the long-run drivers of carbon inequality in OECD economies and highlight the importance of income, trade, and energy structure. These studies establish that carbon inequality is shaped by structural and macroeconomic factors. However, they do not examine whether the composition of technological progress itself is a determinant of carbon inequality. The present paper advances this literature by linking directed technological change to a consumption-based measure of the top-bottom carbon-emissions gap.

The present paper addresses this gap by exploring the association between green technological innovation and intranational carbon inequality in advanced economies. Building on an extension of the directed technical change model of Acemoglu et al. (2012) that incorporates heterogeneous emitters, consumption-based footprints, and rebound effects, we distinguish between two dimensions of technological change: (i) the scale of innovation, proxied by total patent applications, and (ii) the direction of innovation, proxied by the share of environment-related technologies in domestic patenting activity. Carbon inequality is calculated as the logarithm of the absolute emissions gap between the top 10% and the bottom 50% of emitters in OECD economies from 1990 to 2021, based on consumption- and trade-adjusted carbon-footprint data from Chancel (2022) and Nguyen et al. (2025). This measure captures the long-run absolute disparity in

carbon responsibility between high and low emitters, which is the central distributional outcome of interest in this study.

Methodologically, the analysis recognizes that emissions inequality, innovation, and income evolve gradually and may share long-run equilibrium relationships. Accordingly, the empirical strategy employs a set of long-run panel estimators centered on the pooled Bewley approach of Chudik et al. (2026), which is well-suited to heterogeneous macro-panels characterised by cross-sectional dependence and non-stationarity. Compared with the pooled mean group estimator of Pesaran et al. (1999), the pooled Bewley estimator provides a closed-form solution for long-run coefficients and avoids the convergence problems associated with iterative maximum-likelihood procedures while still allowing for heterogeneous short-run dynamics across countries. The baseline findings are then corroborated through a series of robustness checks, using alternative estimators and expanded specifications.

The contributions of this study are threefold. First, it extends the directed technical change literature from aggregate emissions to the distribution of consumption-based carbon footprints. Second, it advances existing OECD carbon-inequality research by examining whether the environmental direction of innovation, beyond income, trade, energy structure, and macroeconomic shocks, is associated with the long-run emissions gap. Third, it distinguishes empirically between aggregate patenting activity and green patent orientation within a cointegrated panel framework, complemented by separate estimations for the emissions of the top 10% and bottom 50% of emitters. In doing so, the paper provides new evidence on whether the composition of technological progress matters for distribution-sensitive sustainability indicators.

These contributions carry direct policy relevance. Understanding who benefits from green innovation is essential for designing equitable and politically sustainable climate policies. If low-carbon technological change reduces emissions inequality, innovation policy can complement carbon pricing and redistributive instruments by narrowing disparities in carbon responsibility. Conversely, if aggregate innovation without environmental orientation fails to alter the emissions gap, targeted interventions may be required to steer technological progress toward socially desirable outcomes. More broadly, the analysis speaks to ongoing debates at the intersection of climate mitigation, innovation policy, and economic inequality.

The remainder of this paper is organized as follows. Section 2 reviews related literature. Section 3 presents the empirical framework and the data. Section 4 reports the econometric results. Section 5 provides robustness checks, and Section 6 concludes the study with policy implications.

2. Related literature

2.1. Carbon inequality and the distribution of emissions

A growing strand of environmental economics emphasizes that carbon responsibility is highly unequal not only across countries but also within them. Using consumption-based footprint accounting, Chancel (2022) shows that top emitters, particularly wealthier groups, account for a disproportionate share of emissions, with within-country disparities playing an increasingly salient role in shaping global carbon inequality. This distributional perspective is consistent with the broader footprint literature documenting how household consumption bundles, especially mobility and housing, drive unequal carbon footprints (Ivanova et al., 2016; Oswald et al., 2020). These studies establish the empirical basis for treating carbon inequality as a distinct outcome rather than as a simple by-product of aggregate emissions.

A second line of research identifies the structural drivers of unequal emissions. Wang and Zhou (2018) examine global CO₂ emissions inequality from a consumption perspective and use index decomposition to attribute changes in inequality to differences in consumption scale and carbon intensity. Bolea et al. (2020) extend this logic through a

globalized, multi-sector perspective, showing that trade-embedded emissions can reshape responsibility by shifting carbon burdens along supply chains. At the national and regional levels, [Fan et al. \(2020\)](#) show that government expenditure is an important correlate of provincial CO₂ inequality in China, suggesting that public finance and state capacity may condition unequal emission outcomes.

A complementary strand links inequality in emissions to inequality in economic resources. [Chen et al. \(2020\)](#) examine the relationship between income inequality and CO₂ emissions for G20 economies, indicating that the inequality–emissions relationship can vary by development stage and institutional context. [Wan et al. \(2022\)](#) identify three transmission channels across a panel of 217 countries: consumption, R&D expenditure, and political freedom. More recently, [Apeti et al. \(2025\)](#) focus explicitly on the connection between wealth inequality and carbon inequality, motivating the view that carbon inequality can reflect not only current income but also wealth-driven access to high-carbon lifestyles, durable consumption, and investment patterns. [Chen et al. \(2024\)](#) further suggest that geopolitical risk increases inequality in CO₂ emissions, measured in terms of the per capita carbon footprint of the top 10% income group. Similarly, [Nguyen et al. \(2025\)](#) employ a pooled Bewley estimator on OECD panels to show that energy dynamics, trade openness, and shifts in structural intensity drive long-run differences in absolute carbon footprints between the top 10% and bottom 50%, while [Zhang et al. \(2024\)](#) document cross-country heterogeneity in emissions among developed economies and draw implications for carbon neutrality strategies. Collectively, these studies indicate that carbon inequality is not merely a descriptive statistic; it is shaped by structural forces, such as income distribution, trade integration, public policy, and consumption composition, that evolve over time.

Simply put, the literature implies that carbon inequality is shaped by persistent structural forces: income and wealth concentration, consumption composition, trade integration, energy structure, and institutional conditions. However, it has paid less attention to whether technological change alters the distribution of emissions across emitters.

2.2. Directed technical change and environmental innovation

Contemporary environmental economics stresses that innovation is not environmentally neutral ([Chen and Lee, 2020](#)). The directed technical change framework formalizes how incentives and policies can reallocate R&D from carbon-intensive (“dirty”) to low-carbon (“clean”) pathways, thereby shifting long-run growth and emissions trajectories ([Acemoglu et al., 2012](#)). This framework is central to the present study because it suggests that the environmental consequences of innovation depend not only on how much innovation occurs, but also on what type of innovation is produced.

Patent data are widely used to measure both the scale and the direction of inventive activity. Although patents are imperfect measures of innovation, they allow technology-field classification, cross-country comparison, and analysis of knowledge diffusion through citations ([Griliches, 1990](#)). Early empirical evidence on induced innovation shows that higher energy prices stimulate energy-saving innovation ([Popp, 2002](#)), while renewable-energy policies are associated with increased patenting in greener technologies ([Johnstone et al., 2010](#)). These findings support the view that market incentives and policy instruments can influence not only technology adoption but also the direction of inventive activity.

Recent studies further demonstrate that innovation is path-dependent and shaped by policy signals. Because firms tend to build on existing knowledge stocks, the dominance of dirty technologies can persist unless policy incentives are sufficiently strong to redirect R&D toward cleaner alternatives ([Noailly and Smeets, 2015](#); [Aghion et al., 2016](#)). Consistent with this argument, market-based climate regulation, such as the EU Emissions Trading System, has been associated with increased low-carbon patenting among regulated firms ([Dechezleprêtre et al., 2023](#)). At the same time, improvements in measurement have

made it possible to distinguish between general inventive scale and environmental orientation. OECD patent indicators and ENV-TECH classifications provide harmonised measures, including the share of environment-related inventions in domestic patenting, which directly captures the direction of innovation rather than its aggregate volume ([Haščić and Migotto, 2015](#)). [Mehmood and Kaewsaeng-on \(2026a\)](#) show that green innovation and trade openness consistently reduce CO₂ emissions across OECD countries, whereas information globalisation and financial depth exhibit weaker and more uneven effects. Related evidence further indicates that renewable energy, financial development, and broader sustainability conditions shape environmental degradation through heterogeneous and non-linear channels, including quantile-specific effects and EKC-type dynamics ([Mehmood & Kaewsaeng-on, 2024, 2026b](#)).

The preceding discussion motivates a key conceptual distinction between aggregate inventive scale and directed green innovation. Aggregate patenting may reflect technological dynamism but is not necessarily environmentally progressive if it reinforces carbon-intensive systems. By contrast, green innovation captures the reallocation of inventive activity toward technologies that reduce energy and carbon intensity or reliance on fossil fuels. This distinction underpins the paper's hypotheses.

2.3. Innovation scale, rebound effects, and why direction matters for inequality

The reason aggregate innovation may not reduce carbon inequality is that technological progress can generate rebound effects. Improvements in energy efficiency lower the effective price of energy services, and can generate both substitution and income effects that increase energy use and offset engineering savings ([Turner, 2013](#); [Greening et al., 2000](#); [Sorrell, 2009](#)). [Borenstein \(2015\)](#) provides a microeconomic framework that decomposes rebound into substitution and income components and clarifies why complete “backfire” is unlikely, while still allowing substantial partial offsets. At the macro level, [Stern \(2020\)](#) shows that general-equilibrium responses through prices, production structure, and induced demand may make economy-wide rebound effects larger than engineering estimates suggest. [Vélez-Henao et al. \(2019\)](#) further caution that common technology proxies in STIRPAT-type models may conflate distinct mechanisms and obscure rebound dynamics.

These insights are particularly relevant for carbon inequality because rebound effects are unlikely to be distributionally neutral. Higher-income households tend to have larger discretionary consumption, higher baseline energy-service demand, and greater access to mobility, housing, appliances, and other carbon-intensive goods. If aggregate innovation raises productivity or reduces the cost of energy services without changing the carbon intensity of the underlying technologies, the resulting consumption expansion may be stronger among high emitters. In that case, the scale of innovation could improve efficiency while leaving the emissions gap unchanged or even widening it.

Directed green innovation may have a more systematic distributional effect. Clean technologies can reduce the carbon intensity of energy supply through renewables and electrification, and lower the energy intensity of consumption through efficient appliances, buildings, and transport systems ([Wurlod and Noailly, 2018](#)). Since high emitters' footprints are often concentrated in technology-sensitive categories such as private mobility, residential energy use, and high-demand electricity services, green innovation may disproportionately reduce emissions at the upper end of the distribution. This mechanism implies that innovation direction, rather than innovation scale alone, is more likely to compress the carbon emissions gap. Rebound theory, therefore, strengthens the empirical case for distinguishing general patenting from green patenting in models of carbon inequality ([Gillingham et al., 2016](#)).

The broader implication is that carbon inequality reflects an interaction between distributional structure and technological direction. Income, trade, urbanisation, and institutional context shape

consumption possibilities and technology diffusion, while green innovation affects the carbon intensity of those possibilities (Khan et al., 2024; Nguyen et al., 2025). The contribution of this study is therefore to integrate directed technical change into the macro-distributional literature on carbon inequality. By separating the scale of innovation from its environmental direction, the study examines whether green technological change is associated with a narrower long-run carbon emissions gap in OECD economies.

3. Methodology and data

In this section, we first present the empirical framework by elaborating on the economic model of Acemoglu et al. (2012). Second, we briefly summarize the panel econometric approaches. Finally, we describe the dataset.

3.1. Empirical approach

To ground the empirical analysis, we adapt the directed technical change framework of Acemoglu et al. (2012) to a setting with heterogeneous emitters and consumption-based carbon footprints. The key intuition is that technological change may affect the emissions gap through two distinct channels. Aggregate innovation may raise productivity and improve efficiency, but it may also expand income, consumption, and demand for energy services. Its distributional effect is therefore ambiguous. By contrast, innovation directed toward low-carbon technologies is more likely to reduce the carbon intensity of energy, transport, housing, and durable consumption, which are important components of high-emitter footprints. This motivates the empirical distinction between the scale of innovation and the environmental direction of innovation.

Consider an economy that produces a final good Y_{it} using clean ($x_{C,it}$) and dirty ($x_{D,it}$) intermediates via a CES technology:

$$Y_{it} = \left[\gamma (A_{it}^C L_{C,it})^{\frac{\sigma-1}{\sigma}} + (1-\gamma) (A_{it}^D L_{D,it})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad (1)$$

where each intermediate is produced linearly from its technology stock and sector-specific labor:

$$x_{C,it} = A_{it}^C L_{C,it}, x_{D,it} = A_{it}^D L_{D,it} \quad (2)$$

Here A_{it}^C and A_{it}^D are the stocks of environment-related (clean) and non-environment-related (dirty) technologies, respectively; $L_{C,it}$ and $L_{D,it}$ are sector-specific labor, $\sigma > 1$ is the elasticity of substitution, and $\gamma \in (0, 1)$. Innovation direction responds to relative profitability: for example, a carbon price (or equivalent policy) shifts R&D effort toward clean technologies, altering the composition of the technology stock without necessarily changing its aggregate scale (Acemoglu et al., 2012, pp. 134–136).

Emissions are consumption-based and group-specific. Let $g \in \{T, B\}$ index the top 10% (high emitters) and bottom 50% (low emitters) of the income distribution, with group emissions E_g given by the identity:

$$E_{g,it} = Y_{g,it} \cdot e_{g,it} \quad (3)$$

where $Y_{g,it}$ is group-specific consumption (or income) and $e_{g,it}$ denotes group-specific emission intensity. The carbon inequality measure follows Chancel (2022)'s definition, as in Nguyen et al. (2025):

$$CO2gap_{it} = \ln(E_{T,it} - E_{B,it}) \quad (4)$$

Substituting the emissions identity gives:

$$CO2gap_{it} = \ln(Y_{T,it} e_{T,it} - Y_{B,it} e_{B,it}) \quad (5)$$

This shows that the gap is jointly determined by differences in economic scale and emissions intensity across groups.

Group-specific emissions intensity is assumed to depend on technological composition and structural factors. Clean technologies reduce emissions intensity more strongly than non-green technologies, while aggregate technological progress may also generate rebound effects by lowering the effective cost of energy services and expanding energy-intensive consumption. This rebound channel may be stronger for the high-income group because its consumption basket contain more transport, housing space, durable goods, and discretionary energy services. A compact representation is:

$$e_{g,it} = \phi_g (A_{it}^C)^{-\eta_C} (A_{it}^D)^{-\eta_D} r_{g,it}$$

where $\eta_C > \eta_D > 0$, and $r_{g,it}$ captures group-specific rebound and income-related effects (see also Online Appendix for further details). This formulation implies that a higher green-technology share should reduce emissions intensity, whereas the effect of aggregate innovation is theoretically ambiguous because efficiency gains may be offset by income, scale, and rebound effects.

To move from the theoretical mechanism to the empirical equation, we impose a long-run proportionality condition in which group-specific income or consumption is related to aggregate income:

$$Y_{T,it} = \lambda Y_{it}, Y_{B,it} = (1 - \lambda) Y_{it}, \lambda \in (0, 1)$$

Substituting this condition into the emissions-gap identity gives:

$$CO2gap_{it} = \ln Y_{it} + \ln[\lambda e_{T,it} - (1 - \lambda) e_{B,it}] \quad (7)$$

Equation (7) shows that the emissions gap depends on aggregate income and group-specific emissions intensities. The latter represents the channel through which the composition of technological progress affects the gap.

The detailed algebra that reparameterises clean and dirty technology stocks into total innovation scale and green innovation share is reported in the Online Appendix. In brief, total technology is written as $A_{it} = A_{it}^C + A_{it}^D$, while the green-technology share is $s_{it} = A_{it}^C / A_{it}$. A first-order approximation of the technology-share component and a second-order approximation of the income-related rebound component yield the following long-run cointegrating equation:

$$CO2gap_{it} = \alpha + \beta_1 \ln A_{it} + \beta_2 s_{it} + \gamma_1 \ln Y_{it} + \gamma_2 (\ln Y_{it})^2 + \delta' Z_{it} + \varepsilon_{it} \quad (8)$$

The expected sign of β_2 is negative because a higher green-technology share should reduce the carbon intensity of consumption. By contrast, the sign of β_1 is ambiguous because aggregate innovation captures both efficiency-improving and consumption-expanding effects. The income terms allow for a non-linear development pattern, while Z_{it} captures structural controls.

Because A_{it}^C and A_{it}^D are unobserved, we proxy $\ln A_{it}$ using patent data: total patent applications in logarithmic form (or $LPAT_{it}$) for scale, and the share of environment-related technologies in domestic patenting ($GINO_{it}$) for green technological orientation, $s_{it} = A_{it}^C / A_{it}$ (Ben Youssef, 2020; Hussain et al., 2022; Dahmani, 2024; Khan et al., 2024; Hašič and Migotto, 2015). Per capita income in logarithms ($LGDP_{it}$) is a standard proxy for $\ln Y_{it}$. Substituting these observables into equation (8) and adding structural controls suggested by Aller et al. (2021) and Ren and Zhu (2025) yields the following estimable specification:

$$CO2gap_{it} = \alpha + \beta_1 LPAT_{it} + \beta_2 GINO_{it} + \gamma_1 LGDP_{it} + \gamma_2 (LGDP_{it})^2 + \delta_1 TROP_{it} + \delta_2 UBAN_{it} + \varepsilon_{it} \quad (9)$$

where $TROP_{it}$ and $UBAN_{it}$ are trade openness and urbanisation rates, respectively. This reduced form is the long-run cointegrating vector estimated by the pooled Bewley estimator (Chudik et al., 2026; Asnani et al., 2025), which is suitable for dynamic panels with non-stationary variables and heterogeneous short-run dynamics. Cross-sectional dependence is addressed through diagnostic testing, bootstrap-based inference, and robustness checks.

The coefficient β_2 is interpreted as the long-run semi-elasticity of the absolute emissions gap with respect to a one-percentage-point increase in the green share of domestic inventions. The coefficient β_1 captures the association between the scale of patenting activity and the emissions gap. This distinction allows the analysis to test whether the direction of innovation matters beyond the aggregate volume of inventive activity.

Two testable hypotheses follow directly from equation (9):

H1. Aggregate innovation, measured by overall patenting activity, does not systematically reduce carbon inequality.

H2. A higher share of environment-related technologies is associated with a reduction in the carbon emissions gap between the top 10% and bottom 50% emitters.

These hypotheses are interpreted as long-run conditional relationships rather than strict causal treatment effects. Reverse causality may arise if countries experiencing changes in emissions structure also adjust their innovation policies or patenting behaviour. The empirical strategy, therefore, combines long-run estimation with robustness checks and uses cautious causal interpretation throughout the analysis.

3.2. Econometric approaches

To investigate the long-run relationship between our variables of interest in equation (9), we employ a comprehensive four-stage econometric framework, as depicted in Fig. 1. This approach accounts for non-stationarity, cross-sectional dependence, and heterogeneity in panel data. Panel cointegration estimators are commonly used for persistent macro-environmental variables that may share long-run relationships, including energy use, public capital, growth, and emissions (see e.g., Binh and Nguyen, 2024; Pham, 2024). We apply this logic to a distributional emissions outcome.

(i) Testing for Cross-Sectional Dependence (CSD)

Before conducting the unit root and cointegration analyses, we must address the potential for cross-sectional dependence. Modern globalized economies often experience common shocks (e.g., oil price volatility and global financial crises) or spatial spillovers, which lead to correlated error terms across units. Ignoring the CSD can lead to severe size distortions and spurious inferences in first-generation tests.

To test for cross-sectional dependence, this study employed Pesaran's (2015, 2021) test, which is based on the average pairwise correlation of residuals across cross-sectional units. Because the present dataset is unbalanced, the relevant test statistic is:

$$CD = \sqrt{\frac{2}{N(N-1)}} \left(\sum_{i=1}^{N-1} \sum_{j=i+1}^N T_{ij}^2 \hat{\rho}_{ij} \right)$$

where $\hat{\rho}_{ij}$ represents the sample estimate of the pairwise correlation coefficient between the residuals of units i and j , T_{ij} is the number of time periods jointly observed for pair (i, j) , and N is the number of cross-sectional units. The null hypothesis of the Pesaran CD test is that the errors are weakly cross-sectionally dependent, implying that the average contemporaneous correlation across units tends to zero as the number of cross-sectional units increases. Under the null hypothesis, the test statistic is asymptotically distributed as a standard normal $CD \sim N(0,1)$. Therefore, a statistically significant CD statistic leads to rejection of the null hypothesis and indicates strong cross-sectional dependence, implying that cross-sectional correlations do not vanish asymptotically but remain systematically present across units.

(ii) Panel Unit Root Testing

The integrated order of the variables determines the choice of the cointegration framework. We distinguish between the "first-

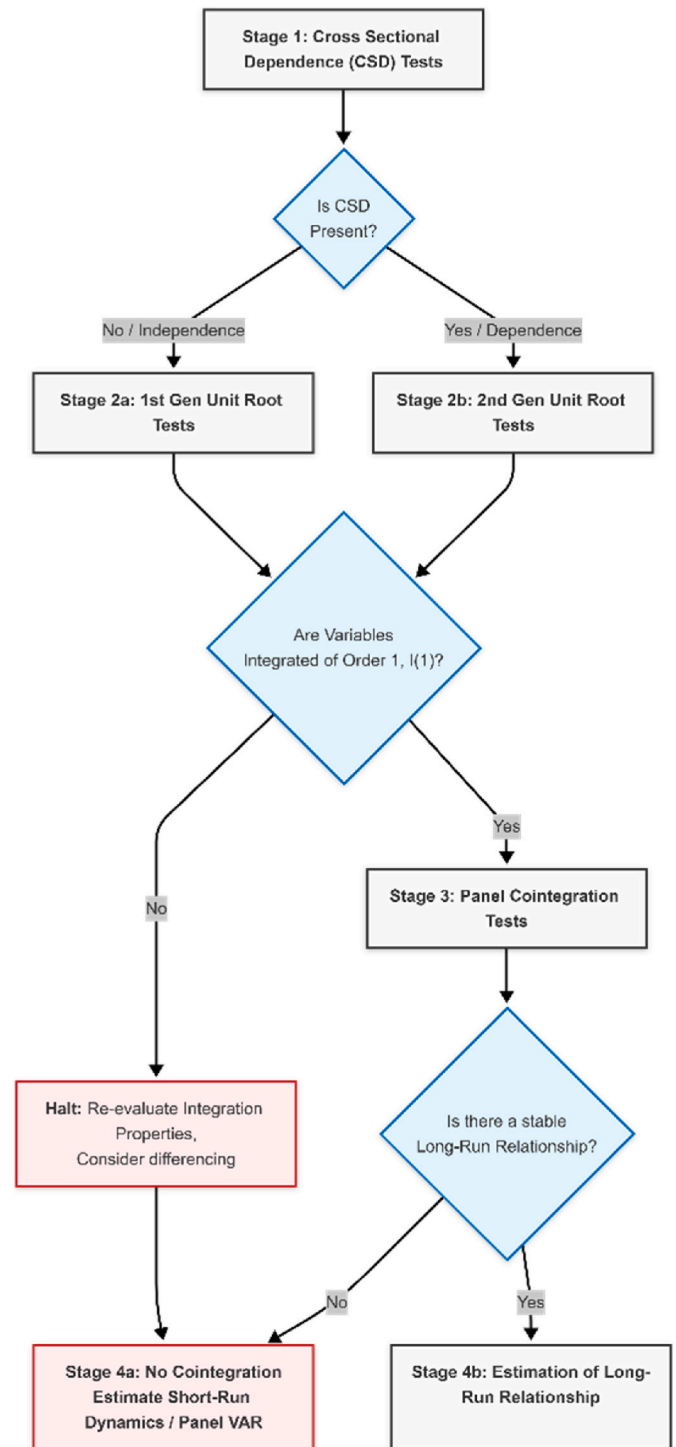


Fig. 1. The empirical workflow.

generation" and "second-generation" tests. For the first type, if the null hypothesis of cross-sectional independence is not rejected, we apply the standard Levin-Lin-Chu (LLC), Im-Pesaran-Shin (IPS), and Breitung tests. The second method involves Pesaran (2007)'s unit root test, which accounts for cross-sectional dependence by adding the cross-sectional averages of lagged levels and first differences of the individual series in the standard ADF regression.

(iii) Panel Cointegration Testing

Once the variables are established to be integrated of order one, $I(1)$, we test for the existence of a long-run equilibrium

relationship using [Westerlund \(2007\)](#)'s error-correction-based panel cointegration tests. These tests are particularly attractive because they do not impose the restrictive assumption—common to residual-based approaches such as [Pedroni \(1999\)](#)—that short-run dynamics and the long-run cointegrating vector are independent. Instead, Westerlund's tests directly evaluate the significance of the error correction (speed-of-adjustment) coefficient α_i in the conditional error-correction model:

$$\Delta y_{it} = d_i - \alpha_i (y_{i,t-1} - \mathbf{b}' \mathbf{x}_{i,t-1}) + \sum_{j=1}^{p-1} \varphi_{ij} \Delta y_{i,t-j} + \delta'_{i0} \Delta \mathbf{x}_{it} + \sum_{j=1}^{p-1} \delta'_{ij} \Delta \mathbf{x}_{i,t-j} + v_{it} \quad (10)$$

Rejection of the null $H_0: \alpha_i = 0$ (no error correction) provides evidence of cointegration. The tests allow for heterogeneous short-run dynamics and are robust to cross-sectional dependence when appropriately augmented.

(iv) Estimation of Long-Run Relationships

To estimate the long-run parameter vector $\mathbf{b}$, we employ the Pooled Bewley (PB) estimator proposed by [Chudik et al. \(2026\)](#) and implemented by [Asnani et al. \(2025\)](#). PB is used as the baseline estimator because our main objective is to estimate a long-run relationship between innovation composition and the carbon emissions gap in a dynamic OECD panel, where short-run adjustment is likely to differ across countries.

The intuition is that observed changes in the emissions gap may reflect both short-run adjustment and long-run equilibrium behaviour. OECD economies differ in energy systems, industrial structures, policy regimes, and innovation capacity, so their short-run responses to green innovation, income, urbanisation, and trade openness are unlikely to be identical. PB accommodates this heterogeneity by allowing country-specific short-run dynamics and adjustment speeds, while pooling the long-run coefficient vector. The common long-run coefficient is therefore interpreted as an average OECD long-run relationship; robustness checks using Mean Group estimates are used to assess whether this restriction is empirically consequential.

Starting from an autoregressive distributed lag (ARDL) model as equation (10), PB applies the Bewley transformation so that the long-run coefficients enter the estimating equation directly:

$$y_{it} = \alpha_i^{-1} d_i + \mathbf{b}' \mathbf{x}_{it} + \psi_i' \Delta \mathbf{z}_{it} + \alpha_i^{-1} v_{it} \quad (11)$$

where y_{it} is the carbon emissions gap, $\mathbf{x}_{it}$ is the vector of long-run regressors, $\mathbf{b}$ is the long-run coefficient vector, α_i is the country-specific adjustment speed, and $\Delta \mathbf{z}_{it}$ contains contemporaneous and lagged first differences of the dependent and explanatory variables. In practical terms, PB estimates the homogeneous cointegrating vector $\mathbf{b}$ after partialling out heterogeneous short-run dynamics. The formal estimator is reported in [Chudik et al. \(2026, equations \(8\)–\(10\)\)](#).

PB is preferred to alternative estimators for several reasons. Unlike the Mean Group estimator, it is designed to estimate a common long-run relationship rather than fully country-specific coefficients. Unlike PMG, it derives the long-run estimates analytically and avoids numerical maximisation, which may be sensitive to convergence problems or local optima. While FMOLS and DOLS provide useful alternatives, this approach is more directly aligned with the dynamic heterogeneous ARDL framework employed here. Monte Carlo evidence in [Chudik et al. \(2026\)](#) also indicates that it can perform favourably in finite samples relative to PMG, panel DOLS, and panel FMOLS, particularly in terms of root mean squared error and size control.

The PB estimator nevertheless has important limitations. It

assumes a stable common long-run coefficient vector and a credible direction of long-run causality from the regressors to the dependent variable. This is restrictive, as reverse causality between green innovation and carbon inequality is plausible: countries with a narrower emissions gap may also exhibit stronger environmental preferences, more inclusive climate policy institutions, or greater support for green technologies. Accordingly, the estimates are interpreted as long-run conditional associations rather than definitive causal effects.

The presence of strong cross-sectional dependence also warrants caution. Combined with sieve wild bootstrap inference (see the Appendix for detailed bootstrap procedure), the approach can yield confidence intervals that are robust to arbitrary cross-sectional dependence in the errors. However, if unobserved common factors are correlated with both green innovation and the emissions gap, cross-sectional dependence becomes an omitted variable problem rather than merely an issue of standard errors. The estimator does not automatically eliminate this source of bias.

We therefore treat the PB estimates as the baseline evidence on long-run conditional associations and evaluate their robustness under alternative modelling assumptions. Mean Group estimates are used to assess whether the imposed common long-run slope conceals substantial cross-country heterogeneity. Fixed-effects models with Driscoll–Kraay standard errors provide a complementary specification that controls for time-invariant country heterogeneity and permits inference under cross-sectional and serial dependence. IV-GMM specifications are employed to examine the sensitivity of the results to potential simultaneity between innovation and the emissions gap, using lagged regressors and predetermined instruments, as motivated by concerns over identification and macro-panel dependence in related emissions models ([Lin and Liscow, 2013](#); [Eibinger et al., 2024](#)). These exercises do not deliver definitive causal identification, but they assess whether the core results are robust to alternative assumptions regarding long-run heterogeneity, error dependence, and endogeneity.

3.3. Data

The empirical analysis uses an unbalanced annual panel of OECD economies over 1990–2021, with country-year coverage varying according to data availability. This period captures the major stages of OECD energy transitions, including the rise in renewables, gains in efficiency, and evolving patterns of inventive activity driven by green policy initiatives. The OECD sample is appropriate for the research question because these countries have relatively long patenting records, more complete environment-related technology indicators, and substantial heterogeneity in innovation capacity, income, energy structure, and carbon-footprint distribution. The starting OECD sample contains 38 economies; the final estimation sample contains 36 after excluding Ireland and Italy because of incomplete data on the share of environment-related technologies in domestic inventions (GINO) ([OECD, 2025, March 12](#)).

Innovation data, patent applications by residents and non-residents (PAT) and the share of environment-related technologies in domestic inventions (GINO), were obtained from the World Bank and OECD Patent Indicators, respectively. The CO₂ inequality metric (CO2gap), defined in equation (4), was calculated using data from the World Inequality Database. We use this log absolute-gap measure, as in [Nguyen et al. \(2025\)](#), because the study focuses on the disparity in per-capita carbon responsibility between high and low emitters. The logarithmic transformation reduces skewness and allows the estimated coefficients to be interpreted approximately as (semi)-elasticities. To assess sensitivity to this construction, the robustness analysis reports separate estimates for emissions from the top 10% and bottom 50% groups, with

the former used in [Chen et al. \(2024\)](#). Additional macroeconomic variables, including real GDP per capita (GDP), trade openness (TROP), and urbanisation (UBAN), were sourced from the World Development Indicators database. The descriptive statistics for the sample are presented in [Table 1](#).

Several data-related limitations should be acknowledged. First, consumption-based emissions estimates are subject to measurement error because they rely on input-output modelling, allocation assumptions, and harmonised international accounts. Second, the unbalanced structure of the panel reflects missing observations for some countries and years, which may limit temporal and cross-country coverage. Finally, patent-based indicators may not be perfectly comparable across countries because patenting propensity, intellectual property regimes, sectoral structures, and national innovation systems differ across OECD economies.

[Fig. 2](#) provides descriptive visual evidence on how emissions from different population groups relate to the direction and scale of technological innovation. Panel A plots the logarithm of per capita CO₂ emissions against the share of environment-related technologies in domestic inventions (GINO). The fitted lines and confidence bands indicate a mild negative association, particularly for the top 10% of emitters, suggesting that a stronger environmental orientation of innovation is associated with lower emissions among high emitters. The top 10% nevertheless remains clearly separated from the bottom 50%, illustrating the persistent emissions gap across country-year observations.

Panel B relates group-specific emissions to the overall scale of inventive activity, measured by the logarithm of total patent applications (LPAT). In contrast to Panel A, both fitted lines are upward-sloping, indicating that aggregate patenting is positively associated with emissions in this bivariate representation. This pattern is consistent with the interpretation that total patenting partly reflects broader economic scale, technological capacity, and demand for energy services, rather than necessarily low-carbon technological change. The vertical distance between the two groups remains pronounced across the patent distribution, suggesting that aggregate innovation is not visually associated with a narrowing of the emissions gap.

4. Results

4.1. Preliminary tests

[Table 2](#) reports three complementary tests for cross-sectional dependence in the panel: the CD test of [Pesaran \(2015, 2021\)](#), the weighted CD test (CDw) of [Juodis and Reese \(2022\)](#), and the bias-corrected statistic (CD*) of [Pesaran and Xie \(2026\)](#).

The results from the standard CD test strongly reject the null hypothesis of weak cross-sectional dependence for all the variables. The statistics are large and highly significant ($p < 0.01$), particularly for the CO₂gap, GINO, LGDP, TROP, and UBAN. This finding provides clear evidence that the variables are affected by common shocks or spillover effects across countries. In macro-environmental settings, such dependence is plausible because countries are simultaneously influenced by

global technological changes, energy markets, and international economic integration. Similarly, the CDw test indicates significant cross-sectional dependence for CO₂gap, GINO, TROP, and UBAN, while the null hypothesis cannot be rejected for LPAT and LGDP, suggesting a weaker dependence for the latter variables after weighting adjustments. The bias-adjusted CD* test yielded mixed evidence, confirming the dependence of LPAT, LGDP, TROP, and UBAN, but not for CO₂gap and GINO. Taken together, these supporting tests broadly stress the presence of cross-sectional dependence in the panel, although their sensitivity varies across the variables. Consequently, these findings justify the use of second-generation panel methods that explicitly account for cross-sectional dependence in subsequent unit root and cointegration analyses.

[Table 3](#) presents the results of first- and second-generation panel unit root tests for the variables in equation (9). The first-generation tests (Levin–Lin–Chu, Breitung, Im–Pesaran–Shin, and ADF–Fisher) assume cross-sectional independence. The reported p-values show that the null hypothesis of a unit root cannot be rejected for most variables in levels at conventional significance levels. In particular, CO₂gap, GINO, LPAT, LGDP, and TROP appear non-stationary across the majority of specifications, while UBAN is stationary only under the Levin–Lin–Chu test.

Given the strong evidence of cross-sectional dependence documented earlier ([Table 2](#)), reliance on first-generation tests alone would risk misleading inferences. We therefore complement them with second-generation procedures that explicitly account for common factors. The [Pesaran \(2007\)](#) CIPS test, estimated with both constant and trend specifications, confirms that the null hypothesis of a unit root cannot be rejected for any variable in levels, as all p-values remain above conventional thresholds.

Overall, the combined evidence from first- and second-generation testing methods indicates that the variables are non-stationary in levels, suggesting they are likely integrated of order one. This supports using panel cointegration techniques in the subsequent empirical analysis to examine long-term relationships among the variables.

The cointegration results, shown in [Table 4](#), provide strong and consistent evidence against the null hypothesis of no long-run relationships among the variables. Under the Kao and Pedroni frameworks, all reported statistics are significant at conventional levels in both the specification without a trend and the specification with a deterministic trend. This suggests that the estimated residuals are stationary and that the CO₂gap, GINO, LPAT, LGDP, LGDP², TROP, and UBAN move together over time in a stable equilibrium relationship.

The evidence is further strengthened by the Westerlund variance ratio test, which is particularly useful in panels affected by cross-sectional dependence. The null hypothesis is rejected in both the no-trend and trend specifications (p-values < 0.05), confirming that the long-run association remains robust, even when cross-sectional dependence is accounted for.

4.2. Long-run estimations

The baseline long-run estimates shown in [Table 5](#) reveal a clear

Table 1
Descriptive statistics, 1990–2021.

Variable	Mean	p50	SD	Min	Max	Obs.
CO2Gap	3.20	3.13	0.48	1.49	5.17	1152
PAT	32120.94	2594.50	94607.31	22.00	621453.00	1128
GINO	10.77	9.99	4.88	0.90	50.00	1110
GDP	30852.66	29464.71	21420.45	3713.18	112417.90	1152
TROP	84.69	70.43	50.16	15.72	397.51	1133
UBAN	76.18	77.41	11.09	47.92	98.12	1152

Note: CO₂gap is the logarithm of the difference between CO₂ emissions per capita of the top 10% and bottom 50% income groups ([Nguyen et al., 2025](#)). GDP is the per capita income in real terms (\$ 2015). TROP and UBAN are measures of trade openness and urbanisation rates, expressed as percentages of GDP and total population, respectively. PAT represents total patent applications by residents and non-residents, but logarithmic transformation (LPAT) is used in the regression analysis to reduce skewness and interpret coefficients as elasticities. GINO is the share of environment-related technologies in domestic inventions (% of domestic inventions).

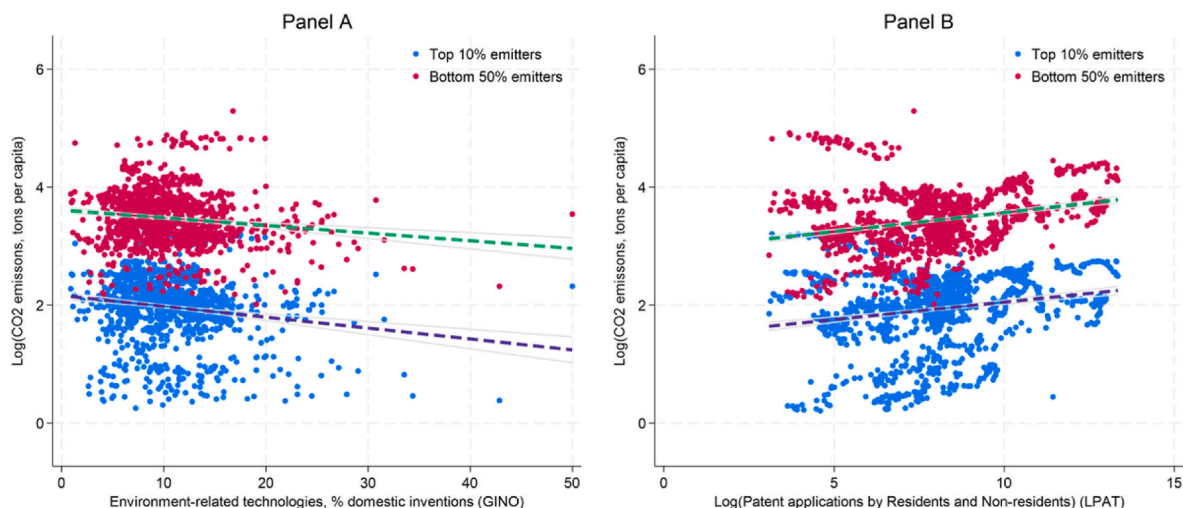


Fig. 2. Carbon inequality and technological innovation in OECD, 1990-2021

Note: Panel A plots the logarithm of per capita CO₂ emissions against the share of environment-related technologies in domestic inventions (GINO); Panel B uses LPAT, the log of total patent applications. Blue and red points denote the top 10% and bottom 50% emitters, respectively. Fitted lines show group-specific bivariate trends with confidence bands. The figure is descriptive and should not be interpreted causally.

Table 2
Cross-sectional dependence (CSD) tests.

	CD	CDW	CD*
CO2gap	33.490 (0.000)	-4.740 (0.000)	0.460 (0.646)
GINO	59.040 (0.000)	-4.570 (0.000)	-0.490 (0.624)
LPAT	3.430 (0.001)	0.720 (0.470)	-2.540 (0.011)
LGDP	155.680 (0.000)	1.100 (0.272)	-3.670 (0.000)
TROP	84.130 (0.000)	-3.680 (0.000)	-2.700 (0.007)
UBAN	74.060 (0.000)	6.070 (0.000)	-3.590 (0.000)

Note: p-values are in parentheses. CD: Pesaran (2015, 2021); CDw: Juodis and Reese (2022); and CD*: Pesaran and Xie (2026). CD and CDw: weak vs. strong cross-sectional dependence; CD*: independence vs. weak dependence in the presence of factors.

Table 3
Panel unit root tests.

	CO2Gap	GINO	LPAT	LGDP	TROP	UBAN
First Generation or Cross-sectional Independent Unit Root Tests (H0: unit root, p-value)						
Levin, Lin & Chu	0.274	0.699	0.993	0.659	0.382	0.007
Breitung	0.977	-	0.120	0.618	-	-
Im, Pesaran & Shin	0.825	0.237	0.765	0.688	0.990	0.999
ADF – Fisher	0.145	0.060	0.355	0.280	0.982	0.896
Second Generation or Cross-sectional Dependent Unit Root Tests (H0: unit root, p-value)						
Pesaran (2007), with constant	0.098	0.421	0.986	0.684	0.105	0.422
Pesaran (2007), with trend	0.480	0.668	0.925	1.000	0.766	0.997

Note: for the 1st gen panel unit root tests, log-level variables are tested with a constant and trend; otherwise, they are tested with a constant.

asymmetry between innovation direction and innovation scale, which closely aligns with the empirical framework in Section 3.1. Across both the pooled Bewley (PB) and pooled mean group (PMG) estimators, GINO has a consistently negative and statistically significant coefficient in all specifications. Because the dependent variable is in logarithms and

Table 4
Panel cointegration tests.

PANEL A: Pedroni				
H0: No cointegration	Without Trend	Trend		
Statistics	Statistics	p-value	Statistics	p-value
Modified Phillips–Perron t	1.717	0.043	2.501	0.006
Phillips–Perron t	-7.901	0.000	-8.992	0.000
Augmented Dickey–Fuller t	-7.471	0.000	-9.170	0.000
PANEL B: Kao				
Modified Dickey–Fuller t	-19.502	0.000		
Dickey–Fuller t	-12.886	0.000		
Augmented Dickey–Fuller t	-4.201	0.000		
PANEL C: Westerlund				
H0: No cointegration	No Trend	Trend		
Variance ratio	-2.227	0.013	-3.160	0.001

Note: Panel cointegration tests are exercised at a BIC-selected lag.

GINO is measured in percentage points, the coefficients are interpreted as semi-elasticities. This indicates that a one-percentage-point increase in the share of environment-related technologies is associated with a reduction in the carbon inequality gap of approximately 0.5–0.6% under PB or 1.0–1.4% under PMG.

The magnitude is economically interpretable. At the sample mean of CO2gap = 3.2, the implied absolute emissions gap is approximately 24.5 tons per capita. The PB estimates, therefore, suggest that a one-percentage-point increase in GINO is associated with an average decline of roughly 0.12–0.15 tons per capita in the absolute gap. A five-percentage-point increase, close to one standard deviation of GINO, corresponds to an approximate 2.5–3.1% decrease in the gap, or about 0.6–0.8 tons per capita at the sample mean. Although the marginal association of a one-percentage-point change is modest, the implied magnitude becomes economically meaningful when changes in green technological orientation are sustained or accumulate over several percentage points. Substantively, these findings support H2 and are consistent with the directed technical change hypothesis: innovation focused on cleaner technologies is associated with a narrower and less unequal distribution of emissions across income groups.

By contrast, the coefficient on LPAT is not statistically significant in any of the PB or PMG specifications reported in Table 5, and its sign is not stable across models. The estimated coefficients are negative in some specifications and positive in others, but none are distinguishable from

Table 5

Baseline long-run estimates using Pooled Bewley and PMG estimators.

VARIABLES	Pooled Bewley				Pooled Mean Group			
CO2gap	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GINO	−0.0063* (0.0017)	−0.0063* (0.0020)	−0.0054* (0.0017)	−0.0051* (0.0017)	−0.0095* (0.0021)	−0.0096* (0.0017)	−0.0131* (0.0018)	−0.0136* (0.0018)
LPAT	−0.0097 (0.0193)	−0.0079 (0.0214)	0.0177 (0.0132)	0.0224 (0.0141)	0.0128 (0.0139)	0.0059 (0.0141)	−0.0123 (0.0139)	−0.0013 (0.0144)
LGDP	2.0319* (0.7629)	1.9797** (0.9373)	1.8706** (0.8120)	2.1675** (0.9014)	1.3149* (0.4933)	2.1005* (0.4776)	1.9504* (0.5130)	1.7948* (0.5095)
LGDP ²	−0.0927** (0.0401)	−0.0885+ (0.0487)	−0.0777+ (0.0420)	−0.0897+ (0.0474)	−0.0598** (0.0259)	−0.1126* (0.0249)	−0.0706* (0.0262)	−0.0731* (0.0261)
TROP		−0.0006 (0.0007)		−0.0004 (0.0005)		0.0034* (0.0006)		0.0018* (0.0005)
UBAN			−0.0152** (0.0060)	−0.0166* (0.0063)			−0.0253* (0.0032)	−0.0192* (0.0030)
ECT					−0.3706* (0.0413)	−0.3511* (0.0415)	−0.3487* (0.0495)	−0.3557* (0.0470)
Hausman MG vs PMG					0.408	0.433	0.200	0.732
Countries	36	36	36	36	36	36	36	36
Observations	1082	1082	1089	1082	1031	1031	1031	1031

Note: (Bootstrapped) standard errors in parentheses. ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$. Two thousand cross-sectional robust bootstrapped replications regarding pooled Bewley estimations (Asnani et al., 2025). PB and PMG denote Pooled Bewley and Pooled Mean Group estimators, respectively.

zero at conventional significance levels. Since LPAT is expressed in logarithms, the coefficient would be interpreted as an elasticity if statistically significant. However, the present results do not provide evidence that aggregate patenting activity systematically is associated with the carbon inequality gap in the long run. This finding is fully consistent with H1 and suggests that the overall scale of innovation, unlike its environmental direction, does not by itself correspond to lower emissions inequality in this OECD sample.

The income variables are also consistent with the extant EKC literature. LGDP is positive and highly significant, whereas LGDP² is negative in most specifications, supporting an inverted-U relationship between economic development and carbon inequality. This pattern suggests that the emissions gap between the top and bottom emitters may widen in the earlier stages of development, but narrow after a sufficiently high income threshold is reached. Among the control variables, UBAN is negative and statistically significant, indicating that greater urbanisation is associated with a smaller carbon inequality gap, potentially through efficiency gains in transport, housing, and energy use in denser urban settings. In contrast, TROP is insignificant in the pooled Bewley estimates but positive and statistically significant in the PMG specifications, suggesting that trade openness may be associated with a wider carbon inequality gap, although this result is less robust across estimators than the coefficients on GINO and UBAN.

The error correction term (ECT) in the PMG estimates is negative and highly significant across all specifications, with coefficients ranging from approximately −0.35 to −0.37. This supports the presence of a stable long-run equilibrium relationship and indicates that approximately 35% to 37% of the short-run disequilibrium is corrected each year.

The supplementary estimates in Table A.1 provide a comparison with alternative dynamic panel estimators. In the MG models, which allow country-specific long-run coefficients, the coefficient on GINO remains negative and significant in the first two specifications but becomes unstable and statistically insignificant once additional controls are included; the income terms are also large and imprecisely estimated. By contrast, the DFE estimates, which impose stronger homogeneity restrictions, consistently show a negative and statistically significant coefficient on GINO, with magnitudes close to those obtained under PMG. The Hausman tests reported in Table 5 and A.1 fail to reject the pooled specifications relative to their fully heterogeneous MG counterparts, lending support to the common long-run slope assumption used in the PB and PMG frameworks. This does not imply that OECD economies are structurally identical, but it suggests that the negative long-run

association between green innovation and the emissions gap is not driven solely by unstable country-specific estimates.

To better understand the distributional mechanism underlying the baseline estimates, we next examine whether innovation is associated with emissions at the top and bottom of the distribution differently. Because the carbon inequality gap is defined as the log difference between the emissions of the top 10% and bottom 50% income groups (emitters), separate regressions for these two groups help to identify which side of the distribution is more closely related to the estimated gap association.

Table 6 shows that environment-related innovation is associated with lower emissions for both groups, as the coefficient on GINO is negative in all specifications and statistically significant in most cases. Quantitatively, a one-percentage-point increase in the share of environment-related technologies is associated with a reduction in emissions of the top 10% (CO2Top10) by approximately 0.49–0.68%, and of the bottom 50% (CO2Bot50) by around 0.55–0.71%. The estimated magnitudes are broadly similar across the two groups, suggesting that a green orientation of technological change is associated with lower emissions at both ends of the distribution, rather than only among higher emitters. In this sense, the baseline relationship between GINO and a smaller absolute emissions gap appears to reflect broadly shared decarbonization tendencies. Because emissions of the top 10% are substantially higher in absolute terms, similar proportional reductions across groups may still imply larger absolute reductions among high emitters, thereby compressing the absolute top–bottom gap.

In contrast, LPAT appears to play a weaker and less consistent role. For the top 10%, the coefficient is positive in all specifications, but is statistically insignificant in most cases, with only weak significance in one model. This suggests that increases in aggregate inventive activity do not systematically correspond to lower emissions among the high emitters. For the bottom 50%, the coefficients are positive and become weakly or strongly significant in the more saturated specifications, implying that greater overall patenting may be associated with slightly higher emissions at the lower end of the distribution.

The supplementary PB estimates in Table A.2 reinforce these distributional findings. Across alternative control sets, GINO remains negative and statistically significant not only for CO2gap, but also for CO2Top10 and CO2Bot50. The coefficients are tightly clustered between −0.0049 and −0.0054, indicating that the estimated association is stable across dependent-variable definitions and model specifications. LPAT is still consistently insignificant, in line with the weaker role of aggregate patenting reported in Table 6. The control variables are also

Table 6

Innovation effects on emissions of the top 10% and bottom 50% emitters.

VARIABLES	Pooled Bewley by Chudik et al. (2026)							
	CO2Top10				CO2Bot50			
GINO	−0.0068** (0.0031)	−0.0050+ (0.0028)	−0.0051+ (0.0027)	−0.0049+ (0.0026)	−0.0071* (0.0019)	−0.0070* (0.0020)	−0.0061* (0.0017)	−0.0055* (0.0018)
LPAT	0.0335 (0.0276)	0.0160 (0.0245)	0.0391 (0.0244)	0.0382+ (0.0219)	−0.0012 (0.0178)	−0.0035 (0.0192)	0.0238+ (0.0130)	0.0268** (0.0134)
LGDP	3.5133* (1.1586)	3.7941* (0.9630)	4.2163* (1.5190)	4.4331* (1.1290)	2.3384* (0.6950)	2.2694* (0.8210)	2.5773* (0.8707)	2.5683* (0.8642)
LGDP ²	−0.1779* (0.0581)	−0.1833* (0.0500)	−0.2040* (0.0734)	−0.2064* (0.0580)	−0.1112* (0.0365)	−0.1046** (0.0425)	−0.1158* (0.0437)	−0.1124** (0.0449)
TROP		−0.0023** (0.0011)		−0.0015 (0.0010)		−0.0010 (0.0007)		−0.0007 (0.0007)
UBAN			−0.0193** (0.0097)	−0.0206** (0.0092)			−0.0162** (0.0065)	−0.0166** (0.0065)
Countries	36	36	36	36	36	36	36	36
Observations	1082	1082	1082	1082	1082	1082	1082	1082

Bootstrapped standard errors in parentheses. **p < 0.01, *p < 0.05, + p < 0.1. CO2Top10 and CO2Bot50 are the logarithms of CO2 emissions from the top 10% and bottom 50% of emitters, respectively. Two thousand cross-sectional robust bootstrapped replications regarding pooled Bewley estimations (Asnani et al., 2025).

broadly consistent with the main estimates: income follows a concave pattern, gross fixed capital formation per capita (LK) is positively associated with emissions outcomes, and urbanisation is consistently negative, whereas trade openness and import to GDP (IMPY) penetration exhibit weaker and less stable relationships.

Taken together, these results reinforce the interpretation that the direction of innovation matters more than its scale for the long-run carbon emissions gap.

5. Robustness checks

In this section, we assess the robustness of the baseline findings using alternative panel estimators and specifications following the guidelines of Eibinger et al. (2024). These checks assess whether the estimated associations between innovation and carbon inequality are sensitive to alternative treatments of heteroscedasticity, serial correlation, cross-sectional dependence, simultaneity, and omitted controls. The robustness analysis serves four purposes. First, fixed-effects models with Driscoll–Kraay (FE–DK) standard errors test whether the results hold under a fixed-effects framework with inference robust to serial correlation, heteroscedasticity, and cross-sectional dependence. Second, IV–GMM specifications assess sensitivity to potential simultaneity between innovation and emissions outcomes. Third, expanded control

specifications evaluate whether the green innovation coefficient remains robust when additional macroeconomic, financial, uncertainty, and institutional variables are included. Fourth, separate estimates for emissions of the top 10% and bottom 50% are reported to determine whether the baseline gap result is more closely related to variation at one or both ends of the distribution.

Table 7 provides additional support for the core findings reported in the previous subsection. Most importantly, GINO remains negative and statistically significant across all specifications under both the FE–DK and IV–GMM, regardless of whether the dependent variable is CO2gap, CO2Top10, or CO2Bot50. These specifications are not intended to replace the long-run cointegration framework; rather, they test the sensitivity of the baseline pattern to alternative assumptions about fixed effects, error dependence, and potential simultaneity. The results indicate that the negative association between environment-related innovation and the emissions gap is not specific to panel cointegration estimators such as PB or PMG. Moreover, GINO is associated with lower emissions at both ends of the distribution. Because the estimated coefficients for CO2Top10 and CO2Bot50 are broadly similar in magnitude, the smaller the absolute gap appears to reflect lower emissions across both groups rather than a sharply asymmetric association concentrated in only one group.

By contrast, the estimates for LPAT remain less stable across

Table 7

Robustness to alternative estimators.

VARIABLES	FE-DK			IV-GMM		
	CO2Gap	CO2Top10	CO2Bot50	CO2Gap	CO2Top10	CO2Bot50
GINO	−0.0044* (0.0012)	−0.0030** (0.0012)	−0.0041* (0.0010)	−0.0166* (0.0030)	−0.0118* (0.0030)	−0.0149* (0.0028)
LPAT	0.0179+ (0.0099)	0.0637* (0.0159)	0.0288* (0.0072)	−0.0008 (0.0103)	0.0541* (0.0083)	0.0130+ (0.0078)
LGDP	1.1742** (0.5473)	2.6946* (0.4831)	1.4827* (0.5123)	1.2720* (0.4700)	3.2209* (0.2962)	1.7073* (0.3898)
LGDP ²	−0.0464 (0.0284)	−0.1270* (0.0253)	−0.0635** (0.0266)	−0.0478** (0.0237)	−0.1465* (0.0160)	−0.0709* (0.0197)
TROP	0.0003 (0.0005)	−0.0012* (0.0002)	−0.0000 (0.0004)	−0.0001 (0.0003)	−0.0019* (0.0002)	−0.0005 (0.0003)
UBAN	−0.0127* (0.0024)	−0.0104* (0.0023)	−0.0124* (0.0022)	−0.0100* (0.0019)	−0.0115* (0.0014)	−0.0103* (0.0018)
constant	−3.0444 (2.7267)	−11.8143* (2.2357)	−4.2238 (2.4992)			
Observations	1082	1082	1082	979	979	979
Hansen J test				0.1780	0.1156	0.2203

Note: Driscoll–Kraay (DK) standard errors in parentheses. **p < 0.01, *p < 0.05, + p < 0.1. FE-DK is fixed-effects with DK standard errors. IV-GMM is fixed-effects instrumental regressions. CO2Top10 and CO2Bot50 are the logarithms of CO2 emissions from the top 10% and bottom 50% of emitters, respectively. All tests are reported p-values. IV-GMM models are instrumented using lagged independent variables and demographic indicators, as suggested by Lin and Liscow (2013).

specifications. Under FE-DK, the coefficient is weakly positive in the CO2gap equation but positive and statistically significant in both group-specific regressions. Under IV-GMM, LPAT remains insignificant in the CO2gap regression, yet is positive and statistically significant for CO2Top10 and weakly significant for CO2Bot50. These results support the interpretation that aggregate patenting does not display a consistent decarbonising profile and may, in some cases, be linked to higher emissions.

Table 8 provides additional robustness checks, augmenting the baseline specification with macroeconomic, financial, and uncertainty-related control variables (Aller et al., 2021). These checks address potential omitted-variable bias by accounting for broader macro-financial and institutional factors that may jointly influence innovation and emissions inequality. The controls include the Financial Development Index (FD) from the IMF, foreign direct investment inflows as a percentage of GDP (FDI), and the inflation rate (INFL), all sourced from the World Bank. In addition, the World Uncertainty Index (WUI), developed by Ahir et al. (2022), is included to capture fluctuations in the global economic environment that may affect emissions dynamics.

The results are broadly consistent with the baseline estimates. In particular, the negative coefficient of GINO remains intact and statistically significant across all specifications under both FE-DK and IV-GMM. This stability indicates that the negative association of environment-related innovation is robust, even when conditioned on other structural factors of emissions and economic activity. Alongside Tables 5–7, the evidence supports the interpretation that green technological orientation is robustly associated with lower carbon-emissions inequality, although the estimates should be read as conditional associations rather than definitive causal effects.

The pattern for LPAT is less stable. In most specifications, the coefficient is statistically insignificant, although one IV-GMM model reports a negative and significant coefficient at the 1% level. Nonetheless, the results remain in line with earlier findings that increases in overall patenting activity do not systematically reduce carbon inequality once broader macroeconomic and institutional conditions are considered.

Among the additional controls, FD enters positively and significantly, implying that more developed financial systems may widen

emissions inequality, potentially by expanding consumption and investment opportunities concentrated among higher-emitting groups. By comparison, INFL and FDI have mixed or limited effects, implying that these broader macroeconomic factors are not central correlates of the carbon inequality dynamics identified in the baseline model. However, WUI is negative and significant, pointing to a narrower emissions gap during periods of heightened economic uncertainty, possibly because downturns constrain consumption of carbon-intensive goods and services. These patterns are pronounced and highly consistent in Table A.3 in the Appendix, even after accounting for additional institutional variables. Appendix Table A.4 further reports dynamic System GMM estimates for reference, showing that the negative association between GINO and the emissions gap remains after including a lagged dependent variable and additional controls.

As an additional check for unobserved common factors, Appendix Table A.5 reports common correlated effects (CCE) estimates. The CCE specification augments the model with cross-sectional averages to proxy latent common shocks shared across countries. The coefficient on GINO remains negative and statistically significant in the CO2gap regression, with a magnitude close to the pooled Bewley baseline. The group-specific CCE estimates are less uniform; therefore, we interpret this evidence conservatively as supplementary support that the main emissions-gap association is not wholly attributable to common shocks.

In sum, the robustness results support the main hypothesis that the environmental direction of innovation is more consistently associated with the absolute carbon-emissions gap than the overall scale of patenting activity. The evidence should nevertheless be interpreted as a robust conditional association rather than as definitive causal identification.

6. Conclusion and policy implications

This study investigated whether technological innovation is associated with lower inequality in carbon emissions across income groups. Building on the framework of directed technical change, the analysis distinguishes between the scale of innovative activity and its technological orientation. Using panel data for OECD countries and a measure

Table 8
Robustness to alternative control variables.

VARIABLESCO2Gap	FE-DK				IV-GMM			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
GINO	−0.0045* (0.0012)	−0.0048* (0.0010)	−0.0044* (0.0011)	−0.0037* (0.0011)	−0.0160* (0.0030)	−0.0170* (0.0026)	−0.0183* (0.0036)	−0.0169* (0.0056)
LPAT	0.0138 (0.0109)	0.0110 (0.0101)	0.0083 (0.0110)	−0.0160 (0.0097)	0.0058 (0.0098)	0.0041 (0.0081)	−0.0032 (0.0081)	−0.0336* (0.0099)
LGDP	1.1590+ (0.5761)	1.6171* (0.4819)	2.0254* (0.5278)	1.8654* (0.5136)	1.6232* (0.3598)	1.9569* (0.2455)	2.5260* (0.1740)	1.7960** (0.8043)
LGDP ²	−0.0448 (0.0291)	−0.0728* (0.0232)	−0.0935* (0.0250)	−0.0814* (0.0237)	−0.0639* (0.0177)	−0.0895* (0.0130)	−0.1176* (0.0099)	−0.0768+ (0.0413)
TROP	0.0002 (0.0005)	0.0004 (0.0005)	−0.0002 (0.0006)	−0.0006 (0.0007)	−0.0002 (0.0003)	0.0003 (0.0003)	−0.0005 (0.0004)	−0.0004 (0.0009)
UBAN	−0.0130* (0.0023)	−0.0140* (0.0025)	−0.0144* (0.0025)	−0.0117* (0.0027)	−0.0090* (0.0020)	−0.0120* (0.0020)	−0.0105* (0.0024)	−0.0074** (0.0034)
INFL	−0.0003 (0.0004)	0.0000 (0.0004)	0.0001 (0.0004)	−0.0001 (0.0004)	0.0006 (0.0009)	0.0010 (0.0007)	0.0013 (0.0009)	−0.0011** (0.0005)
FD		0.2853** (0.1050)	0.3275* (0.0997)	0.2556* (0.0851)		0.4607* (0.0791)	0.5479* (0.1079)	0.5494** (0.2251)
FDI			0.0004 (0.0003)	0.0009 (0.0006)			−0.0044+ (0.0025)	−0.0010 (0.0013)
WUI				−0.0593* (0.0187)				−0.1083* (0.0179)
Observations	1079	1079	1068	999	979	979	970	697
Hansen J test					0.2227	0.2286	0.2975	0.1298

Note: Driscoll–Kraay (DK) standard errors are in parentheses. ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$. FE-DK denotes fixed effects with DK standard errors. IV-GMM models are instrumented using lagged regressors and lagged population dependency ratio, following the identification concerns discussed by Eibinger et al. (2024) and Lin and Liscow (2013). FD is the IMF Financial Development Index. INFL denotes CPI inflation, FDI denotes foreign direct investment inflows as a percentage of GDP, and WUI is the World Uncertainty Index of Ahir et al. (2022).

of carbon emissions inequality derived from Chancel (2022), the empirical results provide consistent evidence that the direction of innovation is systematically linked to the distribution of emissions.

Across a range of econometric specifications, environment-related innovation, captured by the share of green technologies in total domestic inventions, exhibits a robust negative association with the carbon emissions gap between the top 10% and bottom 50% income groups. This finding suggests that technological change oriented toward cleaner production and energy systems is associated with a narrower absolute difference in emissions across population groups. By comparison, the scale of innovation, measured by overall patenting activity, does not display a stable relationship with emissions inequality. In most specifications, aggregate patenting is statistically insignificant in the carbon-gap regressions. However, in some group-specific models, it is positively associated with emissions, particularly among top emitters. This suggests that increases in general inventive activity do not automatically translate into more environmentally equitable outcomes.

These results have important implications for climate change and innovation policies. Policies that stimulate innovation broadly, such as generic R&D subsidies or intellectual property incentives, may accelerate technological progress but do not necessarily correspond to lower emissions inequality. Without environmental direction, innovation may expand economic activity and energy demand in ways that disproportionately benefit or intensify emissions among higher-emitting groups. In contrast, policies that explicitly promote environmentally oriented technological development appear more likely to generate both environmental and distributional gains.

From a policy perspective, the findings support the use of instruments that guide innovation toward low-carbon technologies. Carbon pricing, targeted research funding, and green industrial policies may create incentives for firms to develop and deploy technologies that reduce emissions intensity across sectors. Additionally, public investment in clean energy infrastructure and energy-efficient urban systems may facilitate the diffusion of low-carbon technologies across income groups, thereby contributing to a more balanced distribution of emissions.

The results also have implications for sustainability monitoring. Standard indicators typically emphasise aggregate national emissions, carbon intensity, renewable energy shares, or counts of green technology. While these measures are essential, they do not capture how carbon responsibility is distributed within countries. The top–bottom emissions

gap used in this study offers a complementary, distribution-sensitive indicator, helping assess whether low-carbon technological progress is associated not only with overall decarbonization but also with narrower emissions disparities across population groups. From this perspective, innovation metrics such as GINO should be interpreted alongside distributional footprint measures, rather than as substitutes for deployment- or outcome-based indicators.

This study has several limitations. First, the analysis focuses on OECD economies, where institutional and technological capacities differ from those of developing countries. Second, patent indicators capture the direction of inventive activity, but may not fully reflect the diffusion or adoption of new technologies. Therefore, future research could explore the role of technology deployment, energy transitions, and policy design in shaping carbon inequality across a broader set of countries.

Overall, the findings reinforce a central message: innovation alone is unlikely to ensure equitable decarbonization. What appears to matter most is the direction of technological change.

Ethical approval

Not applicable.

CRediT authorship contribution statement

Trang Thi Thuy Nguyen: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Binh Thai Pham:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A. Supplement Regressions

Table A.1
Supplements to baseline regressions

VARIABLES	MG				DFE			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CO2gap								
GINO	−0.0163* (0.0062)	−0.0189* (0.0059)	−0.0677 (0.0611)	0.0055 (0.0139)	−0.0113* (0.0027)	−0.0118* (0.0027)	−0.0099* (0.0025)	−0.0104* (0.0026)
LPAT	−0.2028 (0.1252)	−0.1241 (0.1216)	−1.0565 (0.6913)	−0.0314 (0.1455)	0.0002 (0.0162)	0.0023 (0.0164)	0.0140 (0.0156)	0.0179 (0.0158)
LGDP	31.4443 (19.8131)	1.5416 (23.3843)	214.4297 (144.4102)	6.7419 (25.5823)	1.2196** (0.6108)	1.3860** (0.6389)	0.9253 (0.5780)	1.1423+ (0.5994)
LGDP ²	−1.5396 (0.9517)	0.0013 (1.1621)	−10.6388 (7.0923)	−0.2860 (1.2963)	−0.0517 (0.0317)	−0.0624+ (0.0332)	−0.0323 (0.0301)	−0.0458 (0.0313)
TROP		0.0046 (0.0028)		−0.0019 (0.0047)		0.0008 (0.0006)		0.0010+ (0.0005)
UBAN			−0.0676 (0.0949)	−0.0226 (0.1092)			−0.0122* (0.0032)	−0.0132* (0.0032)

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Table A.1 (continued)

VARIABLES	MG				DFE			
CO2gap	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ECT	−0.6067* (0.0567)	−0.6790* (0.0606)	−0.7028* (0.0652)	−0.7838* (0.0616)	−0.4634* (0.0274)	−0.4610* (0.0275)	−0.4916* (0.0283)	−0.4905* (0.0283)
Hausman MG vs DFE	0.450	0.335	0.119	0.815				
Countries	36	36	36	36	36	36	36	36
Observations	1031	1031	1031	1031	1031	1031	1031	1031

Note: Standard errors are in parentheses. **p < 0.01, *p < 0.05, + p < 0.1. MG and DFE are Mean Group and Dynamic Fixed Effects estimators, respectively.

Table A.2
Supplements to Pooled Bewley estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	CO2gap	CO2Top10	CO2Bot50	CO2gap	CO2Top10	CO2Bot50
GINO	−0.0049* (0.0019)	−0.0052* (0.0020)	−0.0054** (0.0026)	−0.0052* (0.0020)	−0.0053* (0.0020)	−0.0052** (0.0026)
LPAT	−0.0011 (0.0155)	0.0071 (0.0152)	0.0200 (0.0225)	−0.0005 (0.0152)	0.0074 (0.0150)	0.0200 (0.0228)
LGDP	1.0557 (0.7505)	1.6144** (0.7680)	4.1712* (1.1122)	1.1180 (0.7449)	1.6891** (0.7571)	4.3256* (1.0869)
LGDP ²	−0.0644 (0.0401)	−0.0923** (0.0399)	−0.2090* (0.0555)	−0.0672+ (0.0394)	−0.0959** (0.0392)	−0.2184* (0.0544)
LK	0.3724* (0.0643)	0.3442* (0.0632)	0.1963** (0.0928)	0.3648* (0.0592)	0.3395* (0.0591)	0.2118** (0.0855)
TROP	0.0006 (0.0004)	0.0003 (0.0004)	−0.0008 (0.0010)			
UBAN	−0.0147** (0.0062)	−0.0157** (0.0064)	−0.0215** (0.0087)	−0.0147** (0.0061)	−0.0156** (0.0063)	−0.0212** (0.0087)
IMPY				0.0014+ (0.0008)	0.0007 (0.0009)	−0.0014 (0.0020)
Countries	36	36	36	36	36	36
Observations	1082	1082	1082	1082	1082	1082

Note: (Bootstrapped) standard errors in parentheses. **p < 0.01, *p < 0.05, + p < 0.1. Two thousand cross-sectional robust bootstrapped replications regarding pooled Bewley estimations (Asnani et al., 2025). PB denote Pooled Bewley and Pooled Mean Group estimators, respectively. LK and IMPY are gross fixed capital formation per capita and imports to GDP, respectively.

Table A.3
Supplements to robustness regressions

	Dependent variable: CO2gap					
	(1)	(2)	(3)	(4)	(5)	(6)
GINO	−0.0034* (0.0011)	−0.0034* (0.0011)	−0.0047* (0.0013)	−0.0046* (0.0010)	−0.0039* (0.0012)	−0.0039* (0.0010)
LPAT	0.0175 (0.0141)	0.0112 (0.0152)	0.0231 (0.0147)	0.0201 (0.0147)	0.0138 (0.0145)	0.0202 (0.0155)
FD	0.3193** (0.1421)	0.3004** (0.1350)	0.2094 (0.1409)	0.3033** (0.1424)	0.3102** (0.1446)	0.2848** (0.1360)
FDI	0.0010+ (0.0005)	0.0010 (0.0007)	0.0012** (0.0005)	0.0010+ (0.0005)	0.0009 (0.0007)	0.0011+ (0.0006)
WUI	−0.0618* (0.0196)	−0.0557* (0.0189)	−0.0625* (0.0179)	−0.0618* (0.0189)	−0.0666* (0.0197)	−0.0664* (0.0205)
INFL	0.0012 (0.0011)	0.0014 (0.0011)	0.0009 (0.0011)	0.0007 (0.0011)	0.0010 (0.0010)	0.0014 (0.0011)
LGDP	1.6728** (0.7543)	1.5526** (0.7267)	2.2670* (0.7733)	1.6432+ (0.8073)	1.7690** (0.7864)	1.8601** (0.6951)
LGDP ²	−0.0712+ (0.0352)	−0.0658+ (0.0338)	−0.1048** (0.0374)	−0.0707+ (0.0377)	−0.0745+ (0.0370)	−0.0792** (0.0327)
TROP	−0.0001 (0.0009)	−0.0003 (0.0009)	0.0002 (0.0009)	−0.0002 (0.0009)	−0.0004 (0.0009)	−0.0001 (0.0008)
UBAN	−0.0112* (0.0032)	−0.0118* (0.0034)	−0.0118* (0.0034)	−0.0117* (0.0034)	−0.0129* (0.0035)	−0.0119* (0.0030)
CONCOR	0.1598* (0.0293)					
GOVEFF		0.1806* (0.0313)				
REGQUA			0.2146* (0.0422)			
RULLAW				0.1634* (0.0557)		
POLSTA					0.0253 (0.0270)	

(continued on next page)

Table A.3 (continued)

	Dependent variable: CO2gap					
	(1)	(2)	(3)	(4)	(5)	(6)
VOIACC						0.1595* (0.0258)
_cons	−6.0164 (3.7813)	−5.2646 (3.6455)	−8.5761** (3.7517)	−5.7196 (3.9977)	−6.2808 (3.8988)	−7.0119+ (3.4862)
Observations	739	739	739	739	739	739

Note: Driscoll-Kraay standard errors in parentheses. **p < 0.01, *p < 0.05, + p < 0.1. FD is the financial development index taken from the IMF. INFL denotes CPI inflation rate, and FDI is FDI inflows as % of GDP. Both indicators are from the World Bank. WUI is the World Uncertainty Index by [Ahir et al. \(2022\)](#). CONCOR (corruption control), GOVEFF (government effectiveness), REGQUA (regulatory quality), RULLAW (rule of law), POLSTA (political stability) and VOIACC (voice accountability) are six institutional quality indicators of the World Governance Indicators dataset.

Table A.4
Supplementary dynamic panel robustness estimates

	(1)	(2)	(3)
	CO2gap	CO2Top10	CO2Bot50
CO2gap(-1)	0.6947* (0.1213)		
CO2Top10(-1)		0.8016* (0.0950)	
CO2Bot50(-1)			0.9269* (0.0524)
GINO	−0.0046* (0.0012)	−0.0042* (0.0013)	−0.0036** (0.0018)
LPAT	0.0229** (0.0102)	0.0145** (0.0070)	0.0073+ (0.0039)
FD	−0.0757 (0.0904)	−0.0594 (0.0582)	−0.0276 (0.0326)
FDI	0.0000 (0.0005)	−0.0000 (0.0004)	−0.0004 (0.0003)
WUI	−0.0252** (0.0107)	−0.0195** (0.0088)	−0.0174* (0.0063)
LGDP	0.3152 (0.4373)	0.2003 (0.2942)	0.0761 (0.2038)
LGDP ²	−0.0095 (0.0216)	−0.0059 (0.0138)	−0.0025 (0.0091)
TROP	0.0002 (0.0003)	0.0002 (0.0002)	0.0003** (0.0001)
UBAN	−0.0001 (0.0011)	−0.0003 (0.0007)	−0.0001 (0.0003)
INSQUA	−0.0047 (0.0260)	0.0082 (0.0139)	0.0106 (0.0113)
POPDPN	−0.0021 (0.0023)	−0.0011 (0.0018)	−0.0013 (0.0014)
_CONS	−1.2140 (2.1278)	−0.7124 (1.3820)	−0.3253 (0.9991)
Obs.	739	739	739
AR(1) test, p-value	0.044	0.024	0.0000
AR(2) test, p-value	0.709	0.911	0.945
Hansen J-test, p-value	0.999	0.408	0.347

Note: Robust standard errors are in parentheses. **p < 0.01, *p < 0.05, + p < 0.1. System GMM specifications include a lagged dependent variable to account for persistence in emissions outcomes. The lagged dependent variable is instrumented using collapsed internal GMM-style instruments dated t−2 and deeper. The remaining regressors and year dummies enter as IV-style instruments. AR(1) and AR(2) report p-values for the Arellano–Bond serial-correlation tests. The Hansen test reports the p-value for overidentifying restrictions. INSQUA is the average of six institutional quality indicators of the World Governance Indicators dataset.

Table A.5
Common correlated effects robustness estimates

VARIABLES	Common Correlated Effects (CCE)		
	CO2Gap	CO2Top10	CO2Bot50
GINO	−0.0055** (0.0023)	−0.0035+ (0.0018)	−0.0005 (0.0007)
LPAT	−0.0123 (0.0154)	−0.0162 (0.0128)	−0.0172 (0.0116)
LGDP	10.6071* (1.9814)	10.1155* (1.5716)	7.5755* (1.4135)
LGDP ²	−0.4947*	−0.4720*	−0.3606*

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Table A.5 (continued)

VARIABLES	Common Correlated Effects (CCE)		
	CO2Gap	CO2Top10	CO2Bot50
	(0.0958)	(0.0767)	(0.0763)
TROP	−0.0004	−0.0005	−0.0017**
	(0.0009)	(0.0007)	(0.0007)
UBAN	−0.0496**	−0.0428*	−0.0190 ⁺
	(0.0195)	(0.0162)	(0.0098)
Constant	−36.8443 ⁺	−27.7789	−4.3041
	(22.0126)	(20.0720)	(21.7898)
Countries	36	36	36
Observations	1082	1082	1082

Note: Robust standard errors in parentheses. **p < 0.01, *p < 0.05, + p < 0.1. CO2Top10 and CO2Bot50 are the logarithms of CO2 emissions from the top 10% and bottom 50% of emitters, respectively. The CCE specification includes cross-sectional averages to proxy unobserved common factors that may affect OECD countries simultaneously, such as global technology cycles, energy-price shocks, and macroeconomic disturbances.

B. The PB bootstrapped inference

PB inference is based on the sieve wild bootstrap procedure proposed by Chudik et al. (2026) and implemented by Asnani et al. (2025). After estimating the PB model, the remaining short-run parameters are obtained by least squares and the residuals are used to generate bootstrap innovations through Rademacher random weights. Simulated series are then constructed recursively from the estimated dynamic system, and the PB estimator is re-estimated for each bootstrap sample. The bootstrap bias is computed as the difference between the average bootstrap estimate and the original PB estimate, and the corrected estimator subtracts this estimated bias from the original coefficient. For inference, we use cross-sectionally robust bootstrap confidence intervals, which resample cross-sectionally stacked residual vectors and therefore allow for arbitrary contemporaneous correlation of errors across countries. This procedure improves finite-sample inference under cross-sectional dependence, although it does not by itself remove coefficient bias caused by omitted common factors correlated with the regressors.

The procedure is as follows:

First, estimate the original PB model and obtain the uncorrected long-run coefficient estimate, $\hat{b}$. Given $\hat{b}$, estimate the remaining short-run and adjustment parameters in the error-correction representation by least squares. This produces the residuals from the dependent-variable equation and the regressor equation, usually denoted $\hat{v}_{it}$ and $\hat{u}_{x,it}$.

Second, for each bootstrap replication $r = 1, \dots, R$, generate new residuals by multiplying the estimated residuals by random Rademacher draws:

$$\hat{v}_{it}^{(r)} = e_{it}^{(r)} \hat{v}_{it}, \quad \hat{u}_{x,it}^{(r)} = e_{it}^{(r)} \hat{u}_{x,it}$$

where

$$e_{it}^{(r)} = \begin{cases} -1, & \text{with probability } 1/2 \\ 1, & \text{with probability } 1/2 \end{cases}$$

Using the estimated parameters, the initial values, and these generated residuals, construct simulated bootstrap series $y_{it}^{(r)}$ and $x_{it}^{(r)}$. Re-estimate the PB model on each simulated sample to obtain $\hat{b}^{(r)}$.

Third, estimate the finite-sample bias as the difference between the mean of the bootstrap estimates and the original PB estimate:

$$\widehat{bias}_R = \left[R^{-1} \sum_{r=1}^R \hat{b}^{(r)} - \hat{b} \right]$$

The bootstrap bias-corrected PB estimator is then

$$\hat{b}_R = \hat{b} - \widehat{bias}_R$$

Thus, if the average bootstrap estimate differs systematically from the original estimate, this difference is treated as an estimate of the small-sample bias and subtracted from $\hat{b}$.

Data availability

Data will be made available on request.

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